



Database Management Systems

Week 1, Day 4

MySQL—A RDBMS

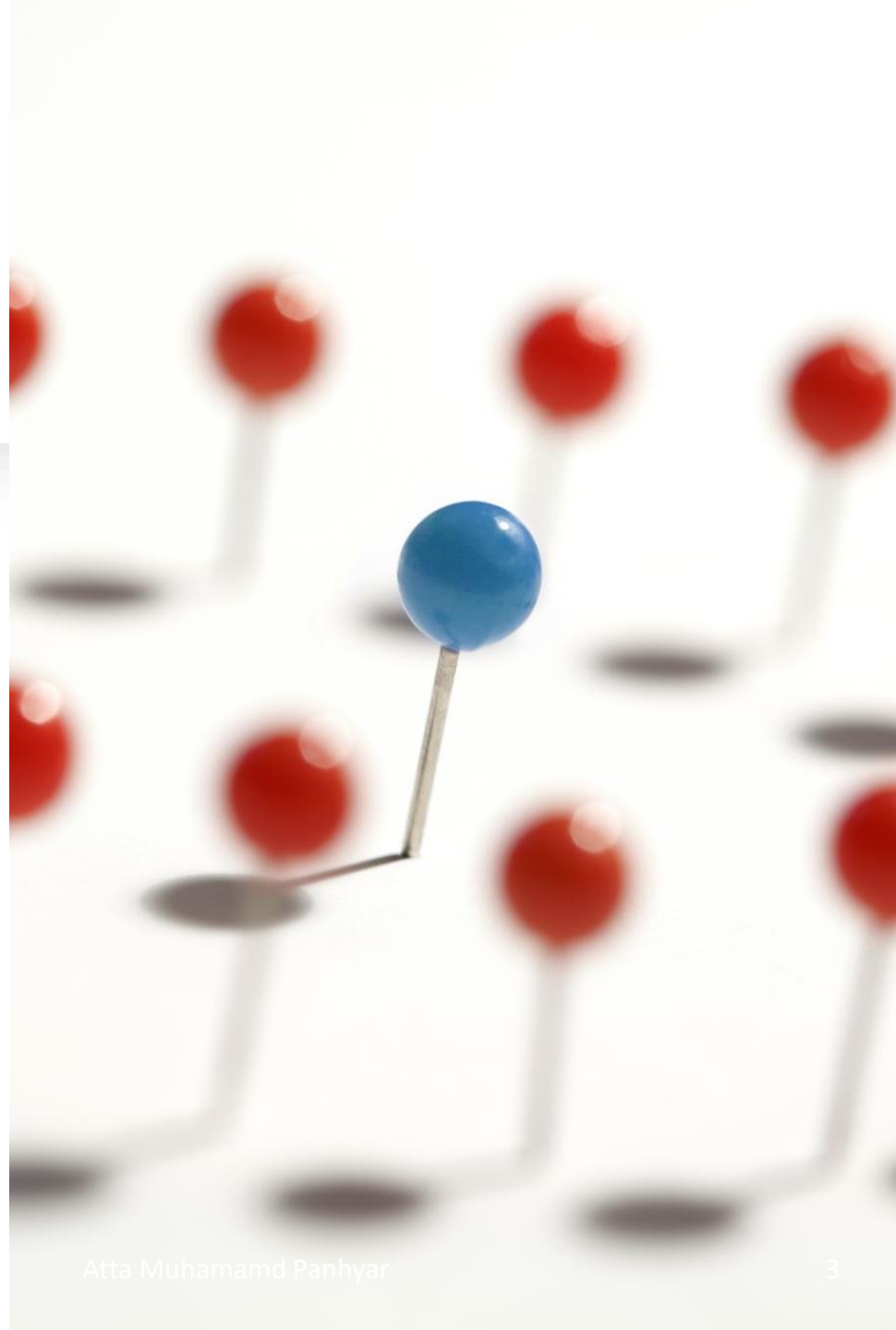
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RECAP

- SELECT
- DISTINCT
- WHERE
- BETWEEN
- IN



ORDER BY

- The **ORDER BY** keyword is used to sort the result-set in ascending or descending order.

```
1 SELECT * FROM Customers
2 ORDER BY Country;
```

Full Grid Filter Rows: Edit: Export/Import: Wrap Cell Content: Fetch rows:					
id	ContactName	Address	City	PostalCode	Country
1	Yvonne Moncada	Ing. Gustavo Mo...	Buenos Aires	1010	Argentina
2	Sergio Gutiérrez	Av. del Libertador...	Buenos Aires	1010	Argentina
3	Patricio Simpson	Cerrito 333	Buenos Aires	1010	Argentina
4	Roland Mendel	Kirchgasse 6	Graz	8010	Austria
5	Georg Pipps	Geislweg 14	Salzburg	5020	Austria
6	Pascale Cartrain	Boulevard Tirou, ...	Charleroi	B-6000	Belgium
7	Catherine Dewey	Rue Joseph-Bens...	Bruxelles	B-1180	Belgium
8	...	...	...	...	...

ORDER BY (ASC/DESC)

- 1 • **SELECT** contactname, address, country **FROM** Customers
- 2 **ORDER BY** Country **DESC**;

contactname	address	country
Carlos Hernández	Carrera 22 con A...	Venezuela
Carlos González	Carrera 52 con A...	Venezuela
Felipe Izquierdo	Ave. 5 de Mayo P...	Venezuela
Manuel Pereira	5ª Ave. Los Palos...	Venezuela
Yoshi Latimer	City Center Plaza ...	USA
John Steel	12 Orchestra Ter...	USA
Rene Phillips	2743 Bering St.	USA
Jaime Yorres	87 Polk St. Suite 5	USA

For several columns

```
1 • SELECT Country, CustomerName FROM Customers
2 ORDER BY Country, CustomerName;
```

Result Grid	
Filter Rows:	Export: Wrap Cell Content: Fetch rows:
Country	CustomerName
Argentina	Cactus Comidas p...
Argentina	Océano Atlántico ...
Argentina	Rancho grande
Austria	Ernst Handel
Austria	Piccolo und mehr
Belgium	Maison Dewey
Belgium	Suprêmes délices
Brazil	Comércio Mineiro

INSERT INTO Statement

- The **INSERT INTO** statement is used to insert new records in a table.

```
INSERT INTO table_name (column1, column2, column3, ...)
VALUES (value1, value2, value3, ...);
```

keep in Mind 🧠



If you are adding values for **all the columns** of the **table**, you do not need to specify the column names in the SQL query.



However, make sure **the order of the values** is in the same order as the columns in the table

INSERT INTO

1 • INSERT INTO

```
2  Customers (CustomerName, ContactName, Address,  
3  City, PostalCode, Country)  
4  VALUES ('Cardinal', 'Tom B. Erichsen', 'Skagen 21',  
5  'Stavanger', '4006', 'Norway');
```

INSERT INTO

```
INSERT INTO Customers (CustomerName, City, Country)  
VALUES ('Cardinal', 'Stavanger', 'Norway');
```

NULL Values

- We will have to use the **IS NULL** and **IS NOT NULL** operators instead

IS NULL

```
1 • SELECT CustomerName, ContactName, Address
2   FROM Customers
3  WHERE Address IS NULL;
```

Grid  Filter Rows: Export:  Wrap Cell Content: 		
CustomerName	ContactName	Address
Cardinal	NULL	NULL

IS NOT NULL

- 1 • **SELECT** CustomerName, ContactName, Address
- 2 **FROM** Customers
- 3 **WHERE** Address **IS NOT NULL**;

Result Grid   Filter Rows: Export:  Wrap Cell Content:  Fetch rows: 			
CustomerName	ContactName	Address	
Alfreds Futterkiste	Maria Anders	Obere Str. 57	
Ana Trujillo Empa...	Ana Trujillo	Avda. de la Const...	
Antonio Moreno ...	Antonio Moreno	Mataderos 2312	
Around the Horn	Thomas Hardy	120 Hanover Sq.	
Berglunds snabbk...	Christina Berglund	Berguvsvägen 8	
Blauer See Delika...	Hanna Moos	Forsterstr. 57	

```
UPDATE Customers  
SET ContactName = 'Alfred Schmidt', City = 'Frankfurt'  
WHERE CustomerID = 1;
```

UPDATE Statement

- The UPDATE statement is used to modify the existing records in a table.

```
UPDATE Customers  
SET PostalCode = 00000  
WHERE Country = 'Mexico';
```

UPDATE Multiple Records

Beware!!

If you forget to include WHERE clause
all record in the table would be
update.

1 **DELETE FROM** Customers **WHERE** CustomerName='Alfreds Futterkiste';

Action Output			
Time	Action	Message	Duration / Fetch
1 09:50:48	UPDATE Customers SET PostalCode = 00000 WHERE Country = 'Mexico'	5 row(s) affected Rows matched: 5 Changed: 5 Warnings: 0	0.125 sec
2 09:51:14	SELECT * FROM customers WHERE country = 'Mexico' LIMIT 0, 10	5 row(s) returned	0.000 sec / 0.000 sec
3 09:54:39	DELETE FROM Customers WHERE CustomerName='Alfreds Futterkiste'	1 row(s) affected	0.234 sec

DELETE Statment

The DELETE statement is used to delete existing records in a table

LIMIT Clause

- The **LIMIT** clause is used to specify the number of records to return.
- The **LIMIT** clause is useful on large tables with thousands of records.
- Returning many records can impact **performance**.

LIMIT EXAMPLE



```
1  SELECT * FROM Customers
2  LIMIT 3;
```

ilt Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:		
CustomerID	CustomerName	ContactName
2	Ana Trujillo Empa...	Ana Trujillo
3	Antonio Moreno ...	Antonio Moreno
4	Around the Horn	Thomas Hardy

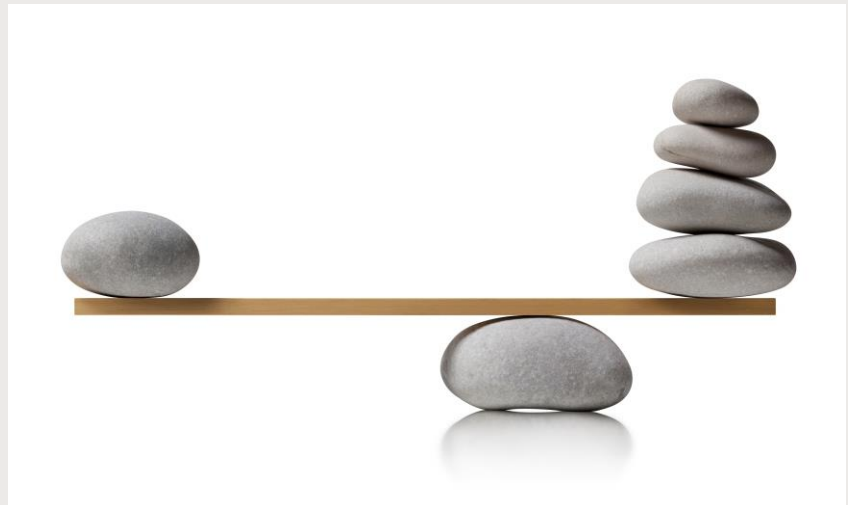
MIN(), MAX() Functions

- The **MIN()** function returns the smallest value of the selected column.
- The **MAX()** function returns the largest value of the selected column.

```
SELECT MIN(Price) AS SmallestPrice  
FROM Products;
```

```
SELECT MAX(Price) AS LargestPrice  
FROM Products;
```

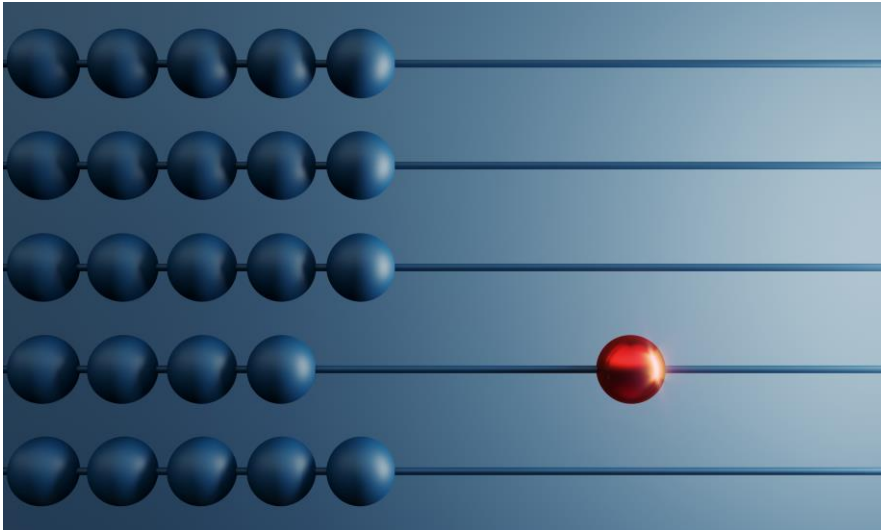
Example



COUNT(), AVG() and SUM()

- The **COUNT()** function returns the number of rows that matches a specified criterion.
- The **AVG()** function returns the average value of a numeric column.
- The **SUM()** function returns the total sum of a numeric column.

Count() Function



```
1 • SELECT COUNT(ProductID)
2 FROM Products;
```

Result Grid		Filter Rows:	Export:	Wrap Cell Content:
COUNT(ProductID)				
77				

```
1 • SELECT AVG(Price)
2   FROM Products;
```

Result Grid		Filter Rows:	Export:	Wr
	AVG(Price)			
▶	28.9221			



AVG() Function


```
1 • SELECT SUM(Quantity)
2   FROM OrderDetails;
```

Result Grid	Filter Rows:	Export:	Wrap Cell Content:
SUM(Quantity)			
12743			

SUM() Function



Question
